

Balanced harvesting in mizer

Size-independent fishing mortality, after Law & Plank (2023)

The three rules

$$\begin{array}{lll} \text{fixed} & F_i(t) & = F_i \\ \text{BH_P} & F_i(t) & = c_P P_i(t) \\ \text{BH_P/B} & F_i(t) & = c_P/B P_i(t) / B_i(t) \end{array}$$

In all three, fishing mortality is independent of body size. Every species enters one mixed fishery at a single entry mass w_f , and all fish above it are caught at the same rate. The balancing is across species, not across sizes. P_i and B_i are the somatic production rate and the biomass over the harvested range.

This run

Model	North Sea (mizer NS_params 3.3.0.9000), 12 species
Entry mass	$w_f = 400$ g, the value used in the paper
Starting state	unfished steady state, then 50 years of fishing
Implementation	$z_i \sim U(0.5, 1.5)$ per species, fixed in time, as in their Fig. 5
Calibration	all three tuned to the same year-50 total yield, 3.28×10^{11} g/yr $F = 0.2 \quad z_i \quad c_P = 1.32 \times 10^{-12} \quad c_P/B = 0.712$
Caution	c_P has dimensions area/mass, so its value belongs to this model's biomass units and does not carry over to another.

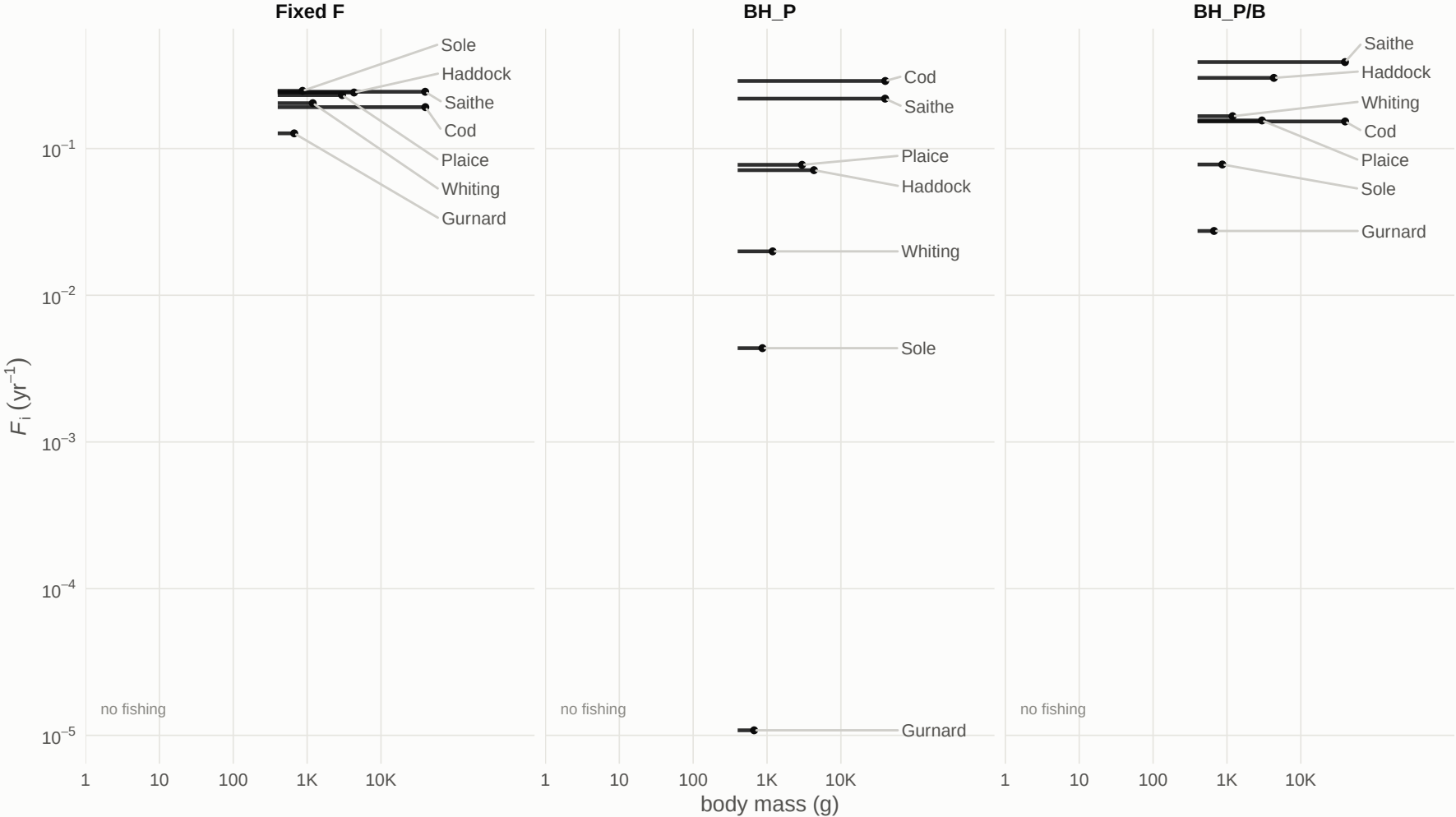
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Law, R. & Plank, M. J. (2023). Fishing for biodiversity by balanced harvesting. Fish and Fisheries.
doi:10.1111/faf.12705 – Generated by bh_run.R and bh_figures.R in this project.

Fishing mortality does not depend on body size

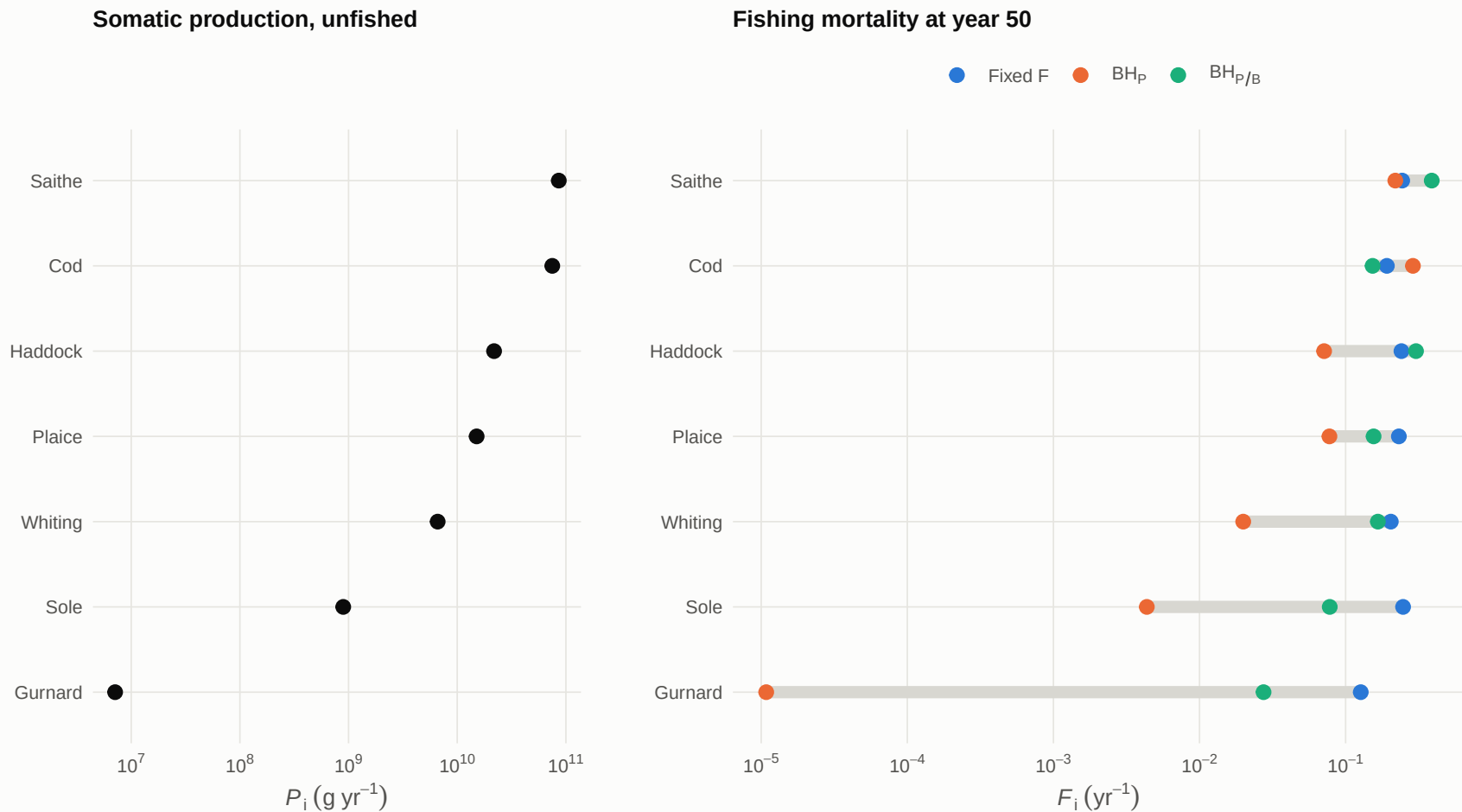
Every species enters one mixed fishery at $w_f = 400$ g, and all fish above that size are caught at the same rate. The three rules differ only in how that one rate is set for each species. Dots mark maximum body size.



North Sea model (mizer NS_params), year 50. Five species with a maximum size below w_f are never caught and are not shown.

Balancing is across species, not across sizes

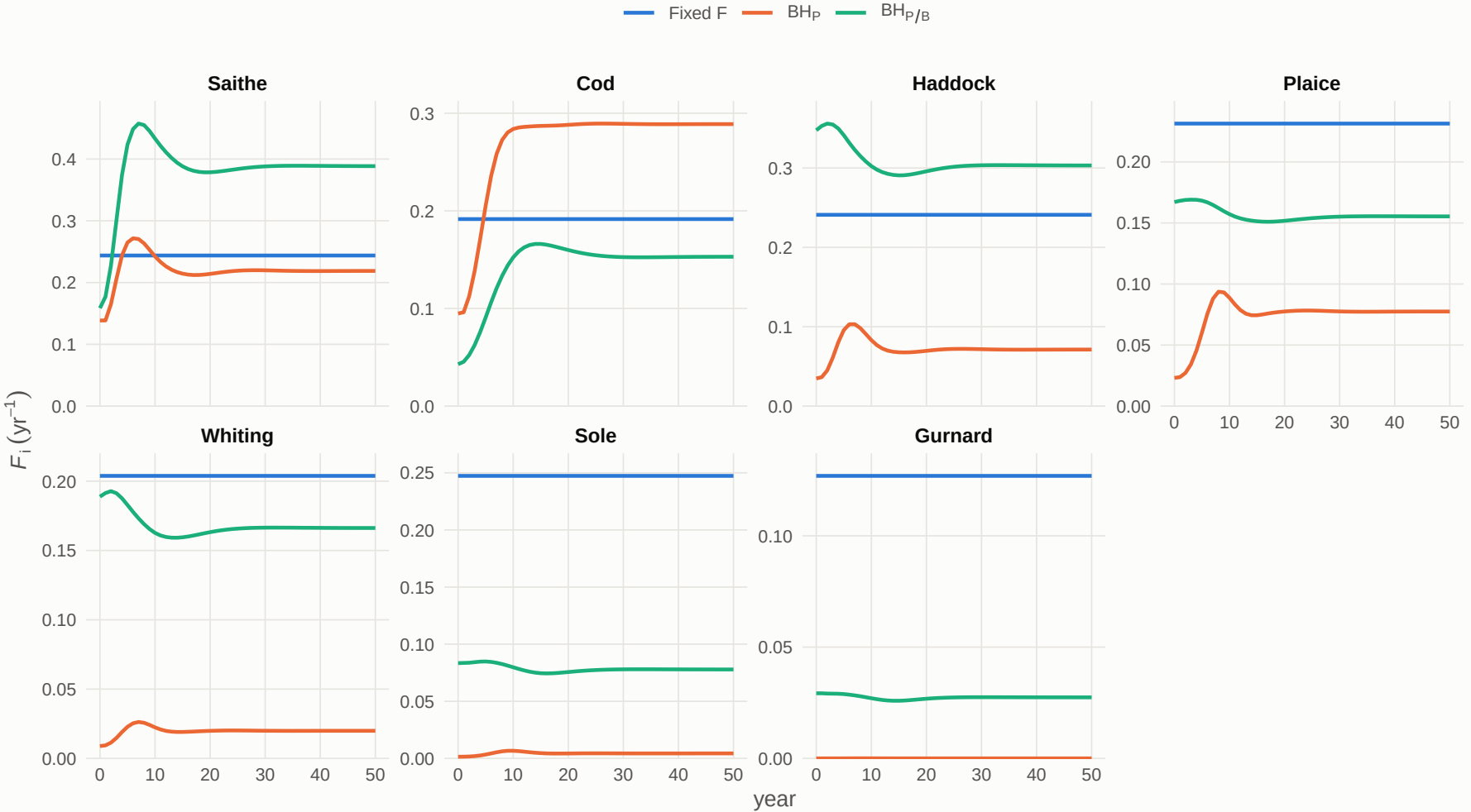
Species ordered by unfished somatic production over the harvested range. BH_P ties F to production, so the least productive species (Gurnard) is fished at $1e-5$ per year – four orders of magnitude below its fixed-F rate.



All three rules are calibrated to the same total yield in year 50.

Under BH_P the fishing mortality is a feedback, not a setting

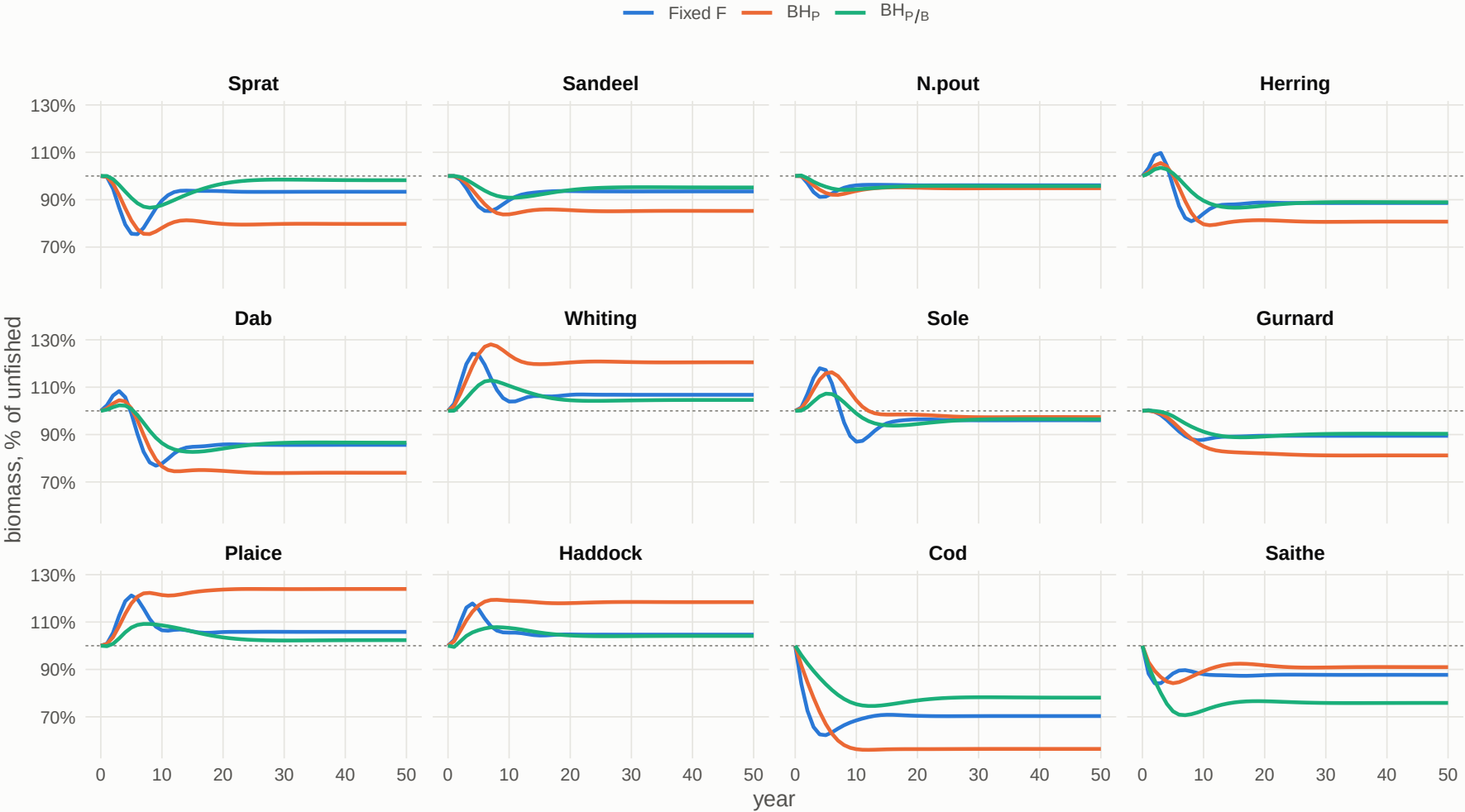
$F_i(t) = c_P P_i(t)$ responds to the stock's own production as fishing changes it. Fixed F is flat by construction; the balanced rules move to a new level over the first decade and then hold it.



Note the independent vertical scales.

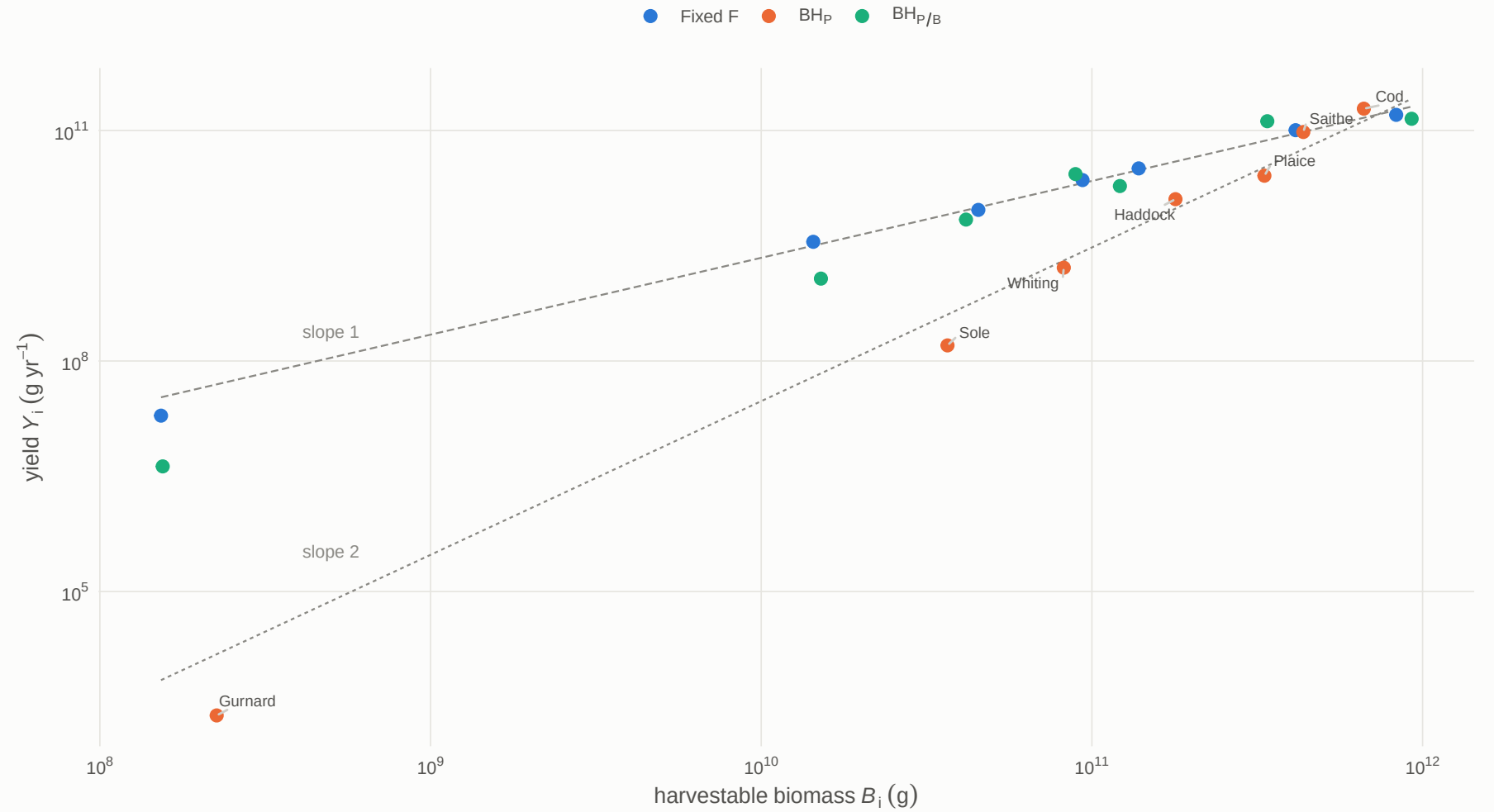
Stock response over 50 years of fishing, at equal total yield

Biomass relative to the unfished state. The first five species are below w_f in maximum size and are never caught directly – what moves them is predation released or intensified by fishing on the others.



Why BH_P takes so little from the unproductive species

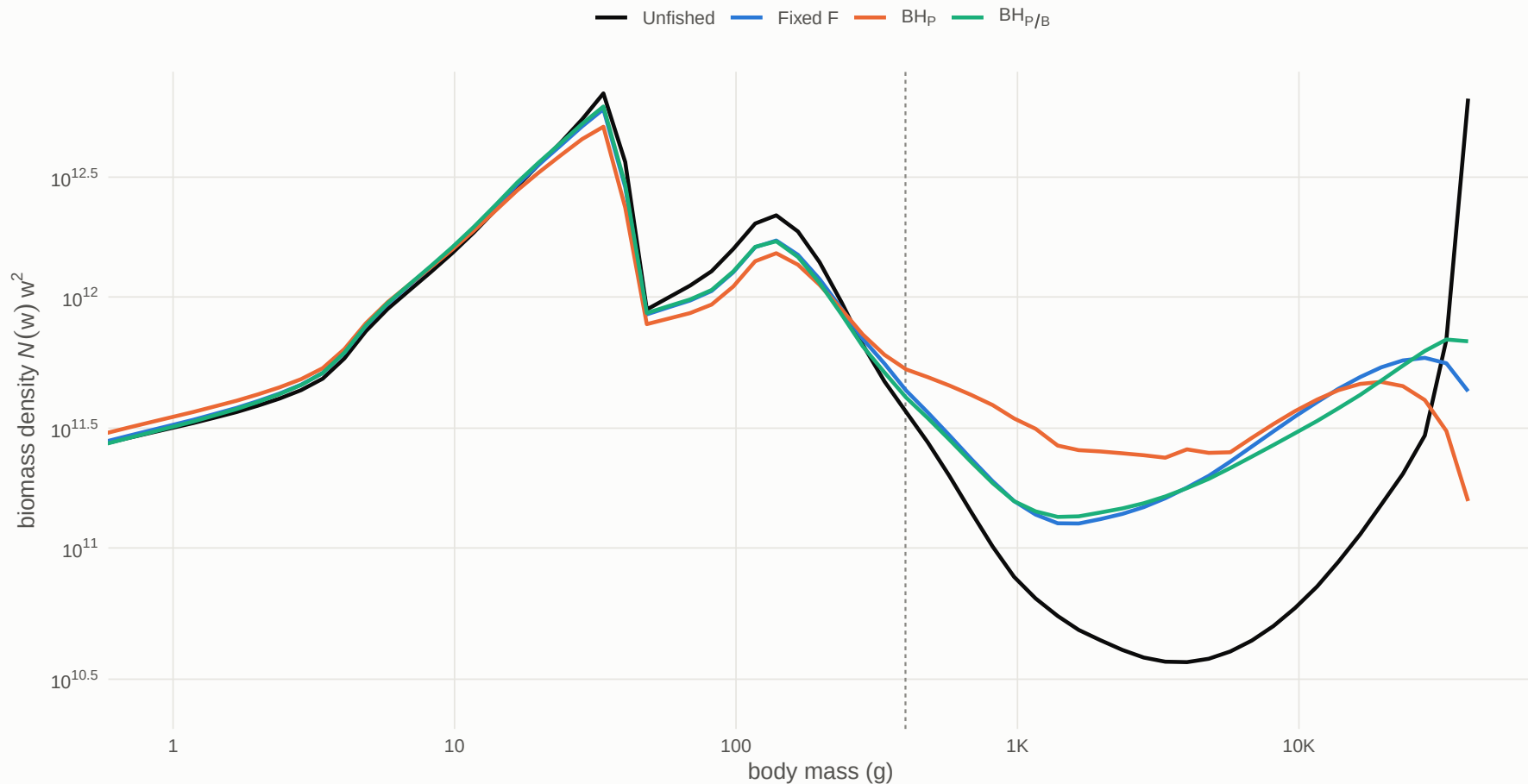
Yield against harvestable biomass in year 50. BH_P makes $Y_i = c_P P_i B_i$, and since production and biomass run together this puts the species near a line of slope 2: halving a species' biomass quarters its catch.



Reference lines are guides, not fits. A fixed F gives slope 1 by construction.

Fishing the big fish lifts the middle of the spectrum

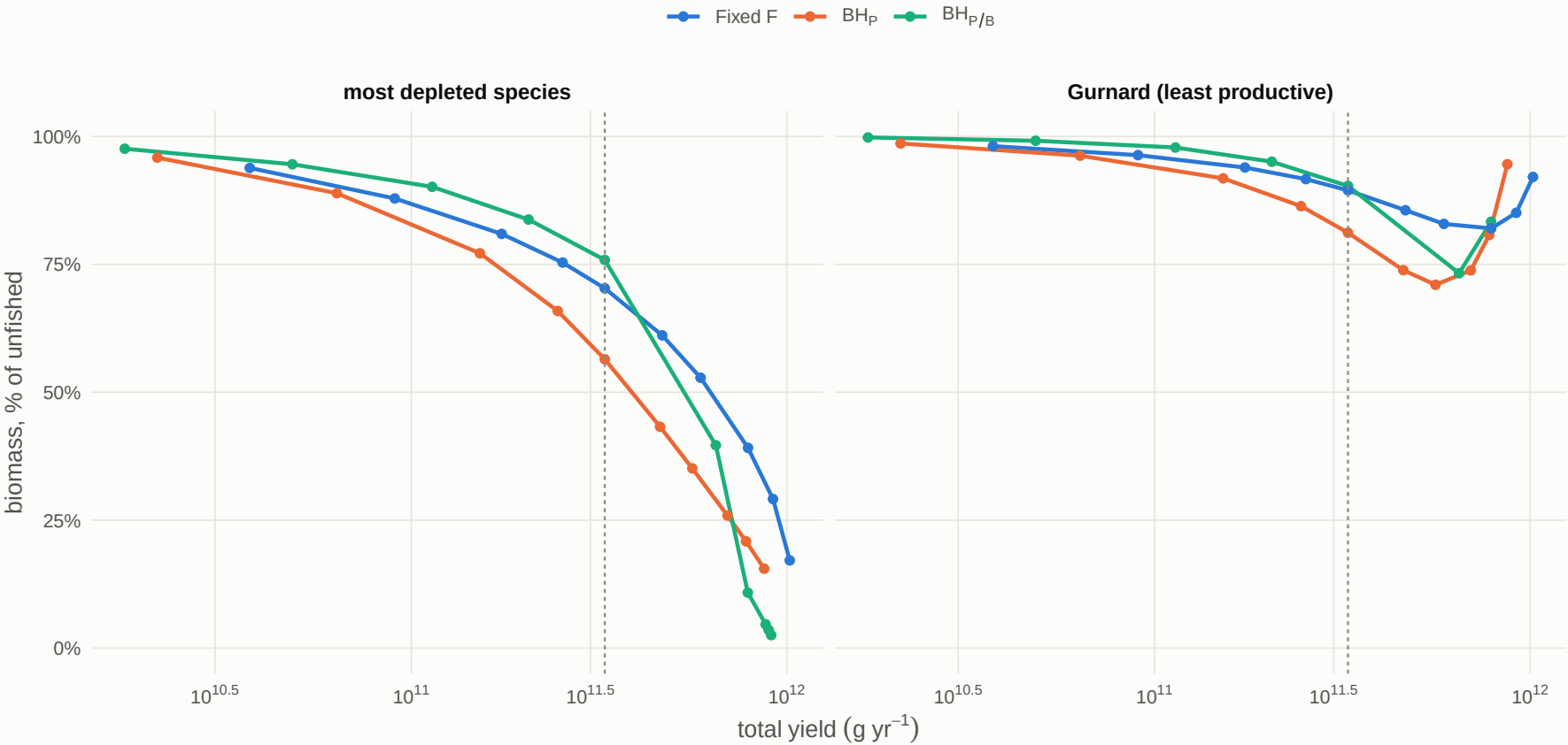
Total fish biomass density across all twelve species, year 50. The unfished state piles biomass up at maximum body size, where growth ceases and only background mortality applies. Removing that pile cuts predation on 1–10 kg fish by more than the fishing it adds – Cod's death rate at 5.7 kg falls from 0.54 to 0.13/yr while F rises by 0.20 – so the middle of the spectrum rises even though total biomass falls.



Dashed line marks $w_f = 400$ g. Recruitment is essentially unchanged in all three regimes.

In this ecosystem, BH_P does not buy any protection

Each rule swept over fishing intensity; the dashed line is the calibrated yield used in the other figures. BH_P fishes Gurnard at $1e-5$ per year – effectively not at all – yet Gurnard ends up slightly worse off than under a fixed F (right). What reaches it is not the hook but the food web. And the species that limits the fishery here is Cod, the most productive one, which BH_P fishes hardest – so BH_P is also the worst of the three on the usual most-depleted test (left).



This is not evidence against Law & Plank. NS_params is not their setting: twelve strongly coupled, well-studied stocks, none rare enough to be at risk of collapse. Their result concerns assembled ecosystems containing genuinely rare species.